

Subgraph tactic	Subgraph timing	What 's important in the logs	What's Important in the tactic
Browser extension			
Initial access	13:40:00 13:40:00	firefox → EXECUTE → pass_mgr (extension triggered repeatedly), firefox → CONNECT → 141.43.176.203:80 (fetch to uncommon IP), firefox → WRITE → gtcache (binary staged in extension dir), pass_mgr → CLONE → /bin/sh, sh → EXECUTE → gtcache, gtcache → CONNECT → 146.153.68.151:80	Correlated evidence of browser or webview fetches to uncommon domains or mutated JS resources (proxy/NGFW logs + Zeek/HTTP logs) followed by unexpected interpreters or script engines executing (python, ruby, sh) spawned from browser processes or user sessions, rapid on-disk staging in /tmp, and outbound connections that deviate from baseline. Defender sees: uncommon resource fetch → short-lived child process executions from user browser context → file writes in temp directories → anomalous outbound C2-like connections.
persistence	13:51:39 13:16:00	sh → CLONE → gtcache gtcache → SENDTO → 146.153.68.151:80 gtcache → OPEN → /var/log/xdev gtcache → WRITE → /var/log/xdev profile → OPEN → /var/log/xdev profile → READ → /var/log/xdev profile → OPEN → /var/log/wdev profile → WRITE → /var/log/wdev profile → UNLINK → /var/log/xdev profile → UNLINK → /var/log/wdev	Detects notify or auditd configuration changes that monitor system files coupled with execution of script interpreters or binaries by cron or systemd timers.
Discovery	13:16:40 13:27:00	profile → WRITE → /var/log/mail /var/log/mail → EXECUTE → /var/log/mail /var/log/mail → CONNECT → 149.52.198.23:80 /var/log/mail → CONNECT → 128.55.12.110_111_22_80 /var/log/mail → CONNECT → 128.55.12.110_111_22_5901_80 /var/log/mail → CONNECT → 128.55.12.110_5901	Detects use of network scanning utilities or scripts performing rapid connections to multiple services or hosts using auditd and netflow/pcap telemetry.
Firefox Backdoor			
initial_access	14:31:22 14:35:16	fluxbox → OPEN → /dev/glx_alsa_675 fluxbox → WRITE → /dev/glx_alsa_675 fluxbox → READ → /dev/glx_alsa_675 fluxbox → CONNECT_RECVFROM_SENDTO → 141.43.176.203:80 fluxbox → CLONE → /usr/bin/firefox firefox → CONNECT_RECVFROM_SENDTO → 146.153.68.151:80	Correlated evidence of browser or webview fetches to uncommon domains... followed by unexpected interpreters or script engines executing spawned from browser processes or user sessions... outbound connections that deviate from baseline. Defender sees: uncommon resource fetch → short-lived child process executions from user browser context → anomalous outbound C2-like connections."
privilege_escalation	14:35:30 14:50:00	14:35:30 clean → EXECUTE → /home/admin/clean 14:35:30 clean → CONNECT → 161.116.88.72:80 14:36:01 clean child → RECVFROM → 161.116.88.72:80 14:36:01 clean child → OPEN/READ → /etc/passwd 14:39:17 clean child → OPEN → / 14:39:38 clean child → OPEN → /root/Documents	Monitor audit logs for setuid/setgid bit changes, executions where UID ≠ EUID (indicative of sudo or privilege escalation), and high-integrity binaries launched by unprivileged users.
Command & Control	14:35:30 14:51:00	14:35:30 clean EXECUTE /home/admin/clean 14:35:30 clean CONNECT_RECVFROM_SENDTO 161.116.88.72:80 14:36:01 clean child RECVFROM 161.116.88.72:80 14:36:01 clean child SENDTO 161.116.88.72:80 14:39:16 clean CLONE child SENDTO 161.116.88.72:80 14:39:17 clean CLONE child SENDTO 161.116.88.72:80	Identifies custom or previously unseen userland processes initiating high-volume HTTP connections with low response volume.
initial_access_2	14:54:16 14:56:32	14:54:16 fluxbox WRITE /dev/glx_alsa_675 14:54:16 fluxbox CONNECT_RECVFROM_SENDTO 141.43.176.203:80 14:54:16 fluxbox CLONE firefox (C21F0F30) 14:54:16 firefox CONNECT_RECVFROM_SENDTO 146.153.68.151:80 14:54:36 firefox child RECVFROM 146.153.68.151:80 14:54:37 firefox child SENDTO 146.153.68.151:80	Correlated evidence of browser or webview fetches to uncommon domains... followed by unexpected interpreters or script engines executing spawned from browser processes or user sessions... outbound connections that deviate from baseline. Defender sees: uncommon resource fetch → short-lived child process executions from user browser context → anomalous outbound C2-like connections."
Persistence	14:56:32 14:58:29	14:56:32 firefox child WRITE /home/admin/profile 14:56:39 profile EXECUTE /home/admin/profile 14:56:39 profile CONNECT_RECVFROM_SENDTO 161.116.88.72:80 14:56:55 profile child RECVFROM 161.116.88.72:80 14:58:29 profile child OPEN /var/log/xdev 14:58:29 profile child WRITE /var/log/xdev 14:58:29 profile child SENDTO 161.116.88.72:80	Detects modification of system or application binaries by monitoring /usr/bin, /bin, and other privileged directories. Correlates file integrity monitoring (FIM) events with unexpected process executions or service restarts.
phishing attachment			
Initial access	14:04:43 14:05:55	fluxbox EVENT_OPEN /tmp/mozilla_admin0/jAG_iSHt.bin.part thunderbird EVENT_OPEN /tmp/mozilla_admin0/jAG_iSHt.bin.part thunderbird EVENT_WRITE /tmp/mozilla_admin0/jAG_iSHt.bin.part x10	Monitor for malicious payload delivery through phishing where attachments or URLs in email clients (e.g., Thunderbird, mutt) result in unusual file creation or outbound network connections. Focus on correlation between mail logs, file writes, and execution activity.
excution	14:06:30 14:06:30	gnome-terminal EVENT_CLONE bash bash EVENT_EXECUTE /tmp/mozilla_admin0/jAG_iSHt.bin.part bash EVENT_OPEN /home/admin/libQt5Widgets.so.5 bash EVENT_OPEN /home/admin/libQt5Gui.so.5 bash EVENT_OPEN /home/admin/libQt5Core.so.5	Cause→effect chain: (1) unified logs show application open/click or crash for Safari/Chrome/Office/Preview/archiver, (2) file write/extraction into ~/Downloads, /private/var/folders/* or ~/Library, (3) parent app spawns osascript/bash/zsh/curl/python or opens a quarantined app with Gatekeeper prompts, (4) network egress from child.

subgraph	tactic	Predicted tactic	edges	Similarity with predicted
discovery_browser_extension	discovery	Persistence	user_process /home/admin/profile open/write log_file /var/log/mail sendto external_host clone user_process /home/admin/profile	user_process /home/admin/ profileopen/write log_file /var/log/xdev sendto external_host clone user_process /home/admin/profile
abstract_execution_phishing_attachment	excution	persistence	gnome-terminal EVENT_CLONE bash bash EVENT_EXECUTE /tmp/mozilla_admin0/jAG_iSHt.bin.part bash EVENT_OPEN /home/admin/libQt5Widgets.so.5 bash EVENT_OPEN /home/admin/libQt5Gui.so.5 bash EVENT_OPEN /home/admin/libQt5Core.so.5	Subgraph is very small and persistence templete have words shell and excute
abstract_privilege_escalation_backdoor	Privilege_Escalation	Persistence	14:35:30 clean → EXECUTE → /home/admin/clean 14:35:30 clean → CONNECT → 161.116.88.72:80 14:36:01 clean child → RECVFROM → 161.116.88.72:80 14:36:01 clean child → OPEN/READ → /etc/passwd 14:39:17 clean child → OPEN → / 14:39:38 clean child → OPEN → /root/Documents	The Privilege Escalation subgraph reads /etc/passwd and browses the /etc/ directory. The Persistence template contains the sentence "use of native tools (e.g., usermod, passwd, groupmod)" — the keyword passwd matches /etc/passwd, and "/etc/profile" and "/etc/init.d/" match the /etc/ directory access. Additionally, the Privilege Escalation template shares two identical sentences with the Persistence template, making both look nearly the same to the encoder (scores: -0.0747 vs -0.0745).
abstract_command_and_control_backdoor	C2	Persistence	14:35:30 clean EXECUTE /home/admin/clean 14:35:30 clean CONNECT_RECVFROM_SENDTO 161.116.88.72:80 14:36:01 clean child RECVFROM 161.116.88.72:80 14:36:01 clean child SENDTO 161.116.88.72:80 14:39:16 clean CLONE child SENDTO 161.116.88.72:80 14:39:17 clean CLONE child SENDTO 161.116.88.72:80	The C&C template content is irrelevant — the problem is that the dominant edge pattern in the C&C subgraph (sendto external_host) was only seen in Persistence during training.